File Processing

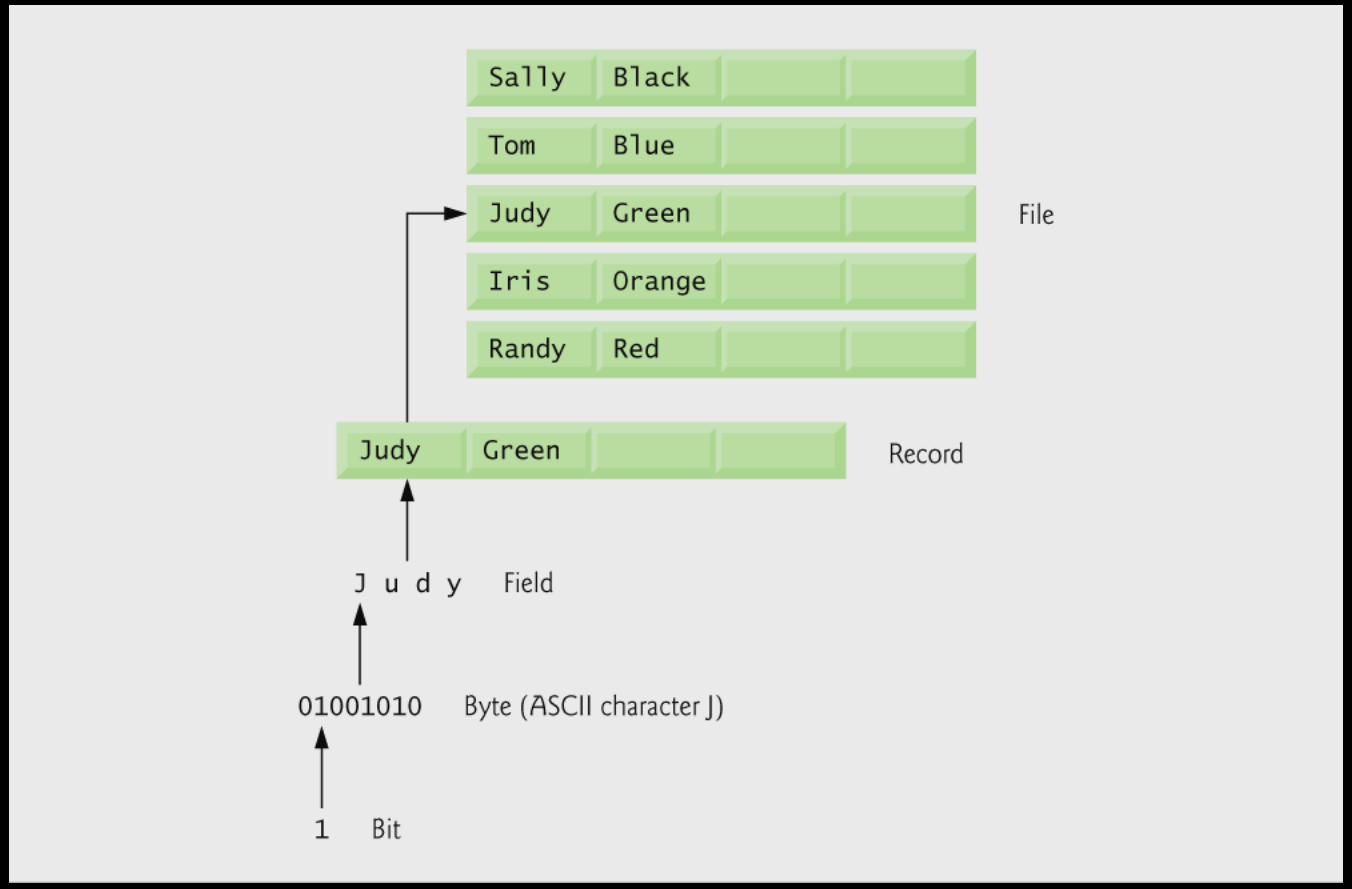
Introduction

- Data files
  - Can be created, updated, and processed by C programs
  - Are used for permanent storage of large amounts of data
    - Storage of data in variables and arrays is only temporary

Data Hierarchy

- Data Hierarchy:
  - Bit – smallest data item
    - `Either value of 0` or `1`
  - Byte – 8 bits
    - Used to store a character
      - Decimal digits, letters, and special symbols
  - Field – group of characters conveying meaning
    - Example: your name
  - Record – group of related fields
    - Represented by a `struct`
    - Example: In a payroll system, a record for a particular employee includes his/her identification number, name, address, etc.
  - File – group of related records
    - Example: a payroll file
  - Database – the group of related files

Data hierarchy



Files and Streams

- C views each file as a sequential stream of bytes
  - File ends with the *end-of-file marker (EOF)*
    - Or, file ends at a specified byte

- When a file is opened, a stream is associated with the file:
  - Provide communication channel between files and programs
  - Opening a file returns a pointer to a `FILE` structure
    - Example file pointers:
      - `stdin` - standard input (keyboard)
      - `stdout` - standard output (screen)
      - `stderr` - standard error (screen)

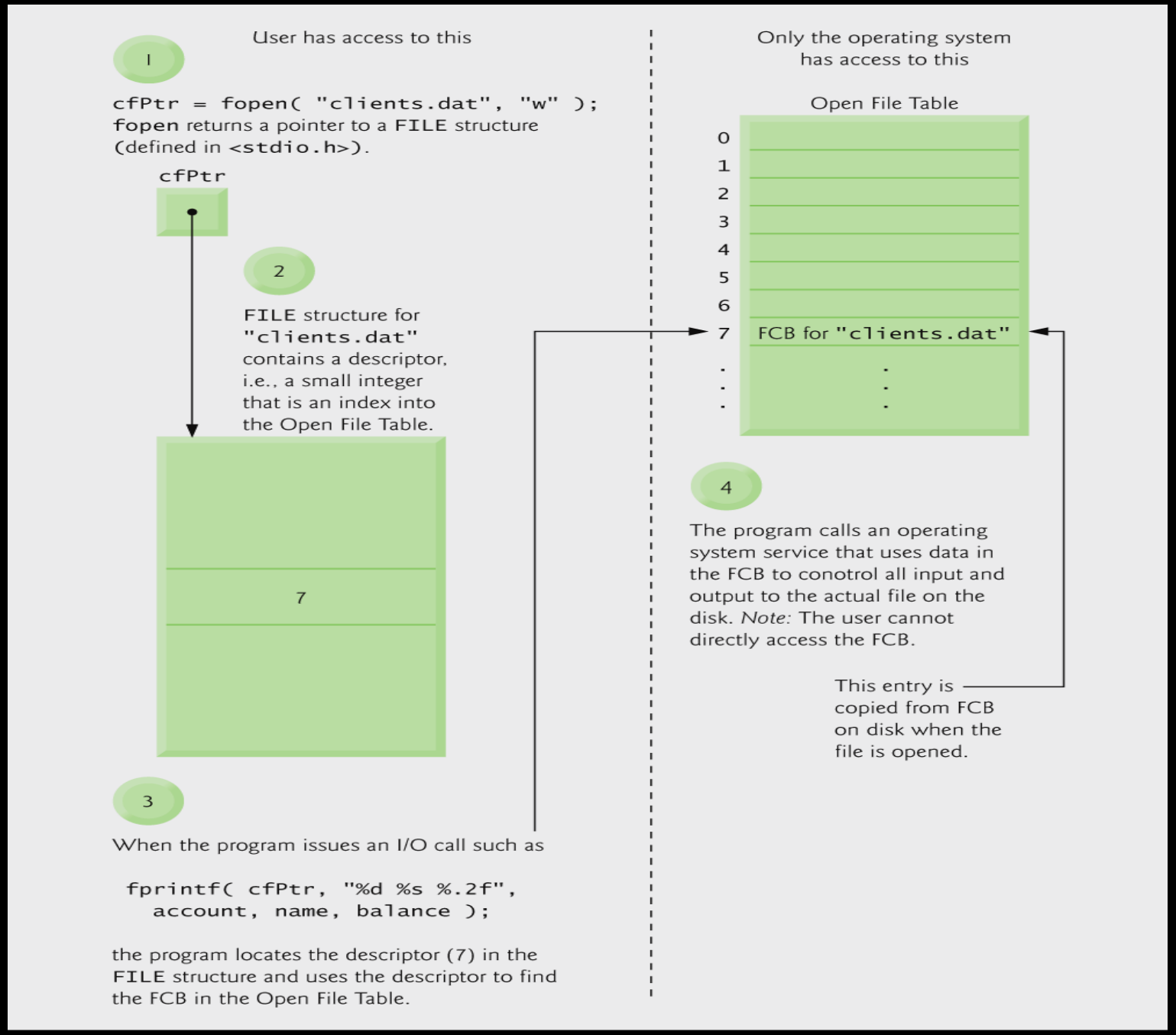
Files and Streams

- `FILE` structure
  - File descriptor
    - A **file descriptor (FD)** is a **small integer (non-negative number)** used by the **operating system** to identify an open file or I/O resource.
    - Index into operating system array called the open file table
  - File Control Block (FCB)
    - system uses it to administer the file

```
struct _iobuf {
   int  _cnt;       // number of bytes left
   char *_ptr;      // next character position
   char *_base;     // location of buffer
   int  _flag;      // mode of file (read/write/error)
   int  _file;      // file descriptor
   int  _charbuf;   // one-char buffer
   int  _bufsiz;    // size of buffer
   char *_tmpfname;  // temporary file name, if any
};
```

```
File Descriptor Table (per process)
---------------------------------
FD 0 → stdin (keyboard)
FD 1 → stdout (screen)
FD 2 → stderr (error output)
FD 3 → myfile.txt  (opened by fopen/fdopen)
FD 4 -> another file
```

Relationship between **FILE** pointers, **FILE** structures and FCBs



Files and Streams

- Read/Write functions in standard library
  - `fgetc`
    - Reads one character from a file
    - Takes a `FILE` pointer as an argument
    - `fgetc( stdin )` equivalent to `getchar()`
  - `fputc`
    - Writes one character to a file
    - Takes a `FILE` pointer and a character to write as an argument
    - `fputc( 'a', stdout )` equivalent to `putchar( 'a' )`
  - `fgets`
    - Reads a line from a file
  - `fputs`
    - Writes a line to a file
  - `fscanf` / `fprintf`
    - File processing equivalents of `scanf` and `printf`

Creating a Sequential-Access File

- C imposes no file structure
  - No notion of records in a file
  - Programmer must provide file structure
- Creating a File
  - `FILE *cfPtr;`
    - Creates a `FILE` pointer called `cfPtr`
  - `cfPtr = fopen("clients.dat", "w");`
    - Function `fopen` returns a `FILE` pointer to file specified
    - Takes two arguments – file to open and file open mode
    - If open fails, `NULL` returned

Creating a Sequential-Access File

- `fprintf`
  - **Used to print to a file**
  - **Like `printf`, except first argument is a `FILE` pointer (pointer to the file you want to print in)**
- `feof( FILE pointer )`
  - **Returns true if end-of-file indicator (no more data to process) is set for the specified file**
- `fclose( FILE pointer )`
  - **Closes specified file**
  - **Performed automatically when program ends**
  - **Good practice to close files explicitly**
- **Details**
  - **Programs may process no files, one file, or many files**
  - **Each file must have a unique name and should have its own pointer**

Creating a Sequential File: Example

```
1  /* Fig. 11.3: fig11_03.c
2     Create a sequential file */
3  #include <stdio.h>
4
5  int main( void )
6  {
7     int account;     /* account number */
8     char name[ 30 ]; /* account name */
9     double balance;  /* account balance */
10
11    FILE *cfPtr;     /* cfPtr = clients.dat file pointer */
12
13    /* fopen opens file. Exit program if unable to create file  */
14    if ( ( cfPtr = fopen( "clients.dat", "w" ) ) == NULL ) {
15       printf( "File could not be opened\n" );
16    } /* end if */
17    else {
18       printf( "Enter the account, name, and balance.\n" );
19       printf( "Enter EOF to end input.\n" );
20       printf( "? " );
21       scanf( "%d%s%lf", &account, name, &balance );
22
```

FILE pointer definition creates new file pointer

fopen function opens a file; **w** argument means the file is opened for writing

Creating a Sequential File: Example

```
23       /* write account, name and balance into file with fprintf */
24       while ( !feof( stdin ) ) {
25          fprintf( cfPtr, "%d %s %.2f\n", account, name, balance );
26          printf( "? " );
27          scanf( "%d%s%lf", &account, name, &balance );
28       } /* end while */
29
30       fclose( cfPtr ); /* fclose closes file */
31    } /* end else */
32
33    return 0; /* indicates successful termination */
34
35 } /* end main */
```

feof returns true when end of file is reached

fprintf writes a string to a file

fclose closes a file

```
Enter the account, name, and balance.
Enter EOF to end input.
? 100 Jones 24.98
? 200 Doe 345.67
? 300 White 0.00
? 400 Stone -42.16
? 500 Rich 224.62
? ^Z
```

End-of-file key combinations for various popular operating systems

Computer system	Key combination
UNIX systems	*<return> <ctrl> d*
IBM PC and compatibles	*<ctrl> z*
Macintosh	*<ctrl> d*

Fig. 11.4 End-of-file key combinations for various popular computer systems.

File opening modes

Mode	Description
`r`	Open an existing file for reading.
`w`	Create a file for writing. If the file already exists, discard the current contents.
`a`	Append; open or create a file for writing at the end of the file.
`r+`	Open an existing file for update (reading and writing).
`w+`	Create a file for update. If the file already exists, discard the current contents.
`a+`	Append: open or create a file for update; writing is done at the end of the file.
`rb`	Open an existing file for reading in binary mode.
`wb`	Create a file for writing in binary mode. If the file already exists, discard the current contents.
`ab`	Append; open or create a file for writing at the end of the file in binary mode.
`rb+`	Open an existing file for update (reading and writing) in binary mode.
`wb+`	Create a file for update in binary mode. If the file already exists, discard the current contents.
`ab+`	Append: open or create a file for update in binary mode; writing is done at the end of the file.

Reading Data from a Sequential-Access File

- Reading a sequential access file
  - Create a `FILE` pointer, link it to the file to read
    ```
    cfPtr = fopen( “clients.dat", "r" );
    ```
  - Use `fscanf` to read from the file
    - Like `scanf`, except first argument is a `FILE` pointer
    ```
    fscanf( cfPtr, "%d%s%f", &account, name, &balance );
    ```
  - Data read from beginning to end
  - File position pointer
    - Indicates number of next byte to be read / written
    - Not really a pointer, but an integer value (specifies byte location)
    - Also called byte offset
  - `rewind( cfPtr )`
    - Repositions file position pointer to beginning of file (byte 0)

```
Get current position :  ftell(FILE *fp)
```

Reading a Sequential File: Example

```
1  /* Fig. 11.7: fig11_07.c
2     Reading and printing a sequential file */
3  #include <stdio.h>
4
5  int main( void )
6  {
7     int account;     /* account number */
8     char name[ 30 ]; /* account name */
9     double balance;  /* account balance */
10
11    FILE *cfPtr;     /* cfPtr = clients.dat file pointer */
12
13    /* fopen opens file; exits program if file cannot be opened */
14    if ( ( cfPtr = fopen( "clients.dat", "r" ) ) == NULL ) {
15       printf( "File could not be opened\n" );
16    } /* end if */
17    else { /* read account, name and balance from file */
18       printf( "%-10s%-13s%s\n", "Account", "Name", "Balance" );
19       fscanf( cfPtr, "%d%s%lf", &account, name, &balance );
20
```

`fopen` function opens a file; **`r`** argument means the file is opened for reading

Reading a Sequential File: Example

```
21        /* while not end of file */
22        while ( !feof( cfPtr ) ) {
23           printf( "%-10d%-13s%7.2f\n", account, name, balance );
24           fscanf( cfPtr, "%d%s%lf", &account, name, &balance );
25        } /* end while */
26
27        fclose( cfPtr ); /* fclose closes the file */
28     } /* end else */
29
30     return 0; /* indicates successful termination */
31
32  } /* end main */
```

`fscanf` function reads a string from a file

```
Account   Name          Balance
100       Jones           24.98
200       Doe            345.67
300       White            0.00
400       Stone          -42.16
500       Rich           224.62
```

Credit Inquiry: Example

```
1  /* Fig. 11.8: fig11_08.c
2     Credit inquiry program */
3  #include <stdio.h>
4
5  /* function main begins program execution */
6  int main( void )
7  {
8     int request;     /* request number */
9     int account;     /* account number */
10    double balance;  /* account balance */
11    char name[ 30 ]; /* account name */
12    FILE *cfPtr;     /* clients.dat file pointer */
13
14    /* fopen opens the file; exits program if file cannot be opened */
15    if ( ( cfPtr = fopen( "clients.dat", "r" ) ) == NULL ) {
16       printf( "File could not be opened\n" );
17    } /* end if */
18    else {
19
20       /* display request options */
21       printf( "Enter request\n"
22          " 1 - List accounts with zero balances\n"
23          " 2 - List accounts with credit balances\n"
24          " 3 - List accounts with debit balances\n"
25          " 4 - End of run\n? " );
26       scanf( "%d", &request );
27
```

Credit Inquiry: Example

```
28       /* process user's request */
29       while ( request != 4 ) {
30
31          /* read account, name and balance from file */
32          fscanf( cfPtr, "%d%s%lf", &account, name, &balance );
33
34          switch ( request ) {
35
36             case 1:
37                printf( "\nAccounts with zero balances:\n" );
38
39                /* read file contents (until eof) */
40                while ( !feof( cfPtr ) ) {
41
42                   if ( balance == 0 ) {
43                      printf( "%-10d%-13s%7.2f\n",
44                         account, name, balance );
45                   } /* end if */
46
47                   /* read account, name and balance from file */
48                   fscanf( cfPtr, "%d%s%lf",
49                      &account, name, &balance );
50                } /* end while */
51
52                break;
53
```

Credit Inquiry: Example

```
54             case 2:
55                printf( "\nAccounts with credit balances:\n" );
56
57                /* read file contents (until eof) */
58                while ( !feof( cfPtr ) ) {
59
60                   if ( balance < 0 ) {
61                      printf( "%-10d%-13s%7.2f\n",
62                         account, name, balance );
63                   } /* end if */
64
65                   /* read account, name and balance from file */
66                   fscanf( cfPtr, "%d%s%lf",
67                      &account, name, &balance );
68                } /* end while */
69
70                break;
71
72             case 3:
73                printf( "\nAccounts with debit balances:\n" );
74
75                /* read file contents (until eof) */
76                while ( !feof( cfPtr ) ) {
77
78                   if ( balance > 0 ) {
79                      printf( "%-10d%-13s%7.2f\n",
80                         account, name, balance );
81                   } /* end if */
82
```

Credit Inquiry: Example

```
83                      /* read account, name and balance from file */
84                      fscanf( cfPtr, "%d%s%lf",
85                         &account, name, &balance );
86                   } /* end while */
87
88                   break;
89
90            } /* end switch */
91
92            rewind( cfPtr ); /* return cfPtr to beginning of file */
93
94            printf( "\n? " );
95            scanf( "%d", &request );
96         } /* end while */
97
98         printf( "End of run.\n" );
99         fclose( cfPtr ); /* fclose closes the file */
100     } /* end else */
101
102     return 0; /* indicates successful termination */
103
104 } /* end main */
```

rewind function moves the file position pointer back to the beginning of the file

Credit Inquiry: Example

```
Enter request
 1 - List accounts with zero balances
 2 - List accounts with credit balances
 3 - List accounts with debit balances
 4 - End of run
? 1

Accounts with zero balances:
300       White           0.00

? 2

Accounts with credit balances:
400       Stone         -42.16

? 3

Accounts with debit balances:
100       Jones          24.98
200       Doe           345.67
500       Rich          224.62

? 4
End of run.
```

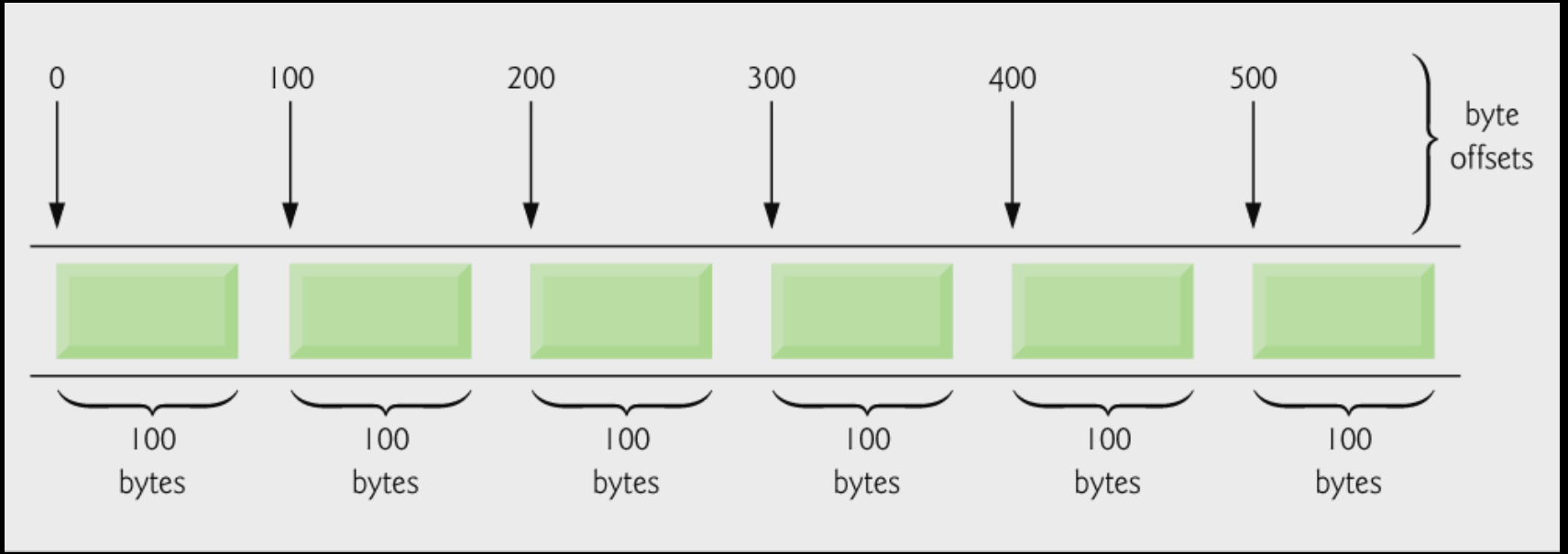
Updating a Sequential-Access File

- Sequential access file
  - Cannot be modified without the risk of destroying other data
    - Example:
      - 300 White 0.00
      - 300 Worthington 0.00
      - Characters beyond second ‘o’ will overwrite the beginning of the next sequential record
  - Fields can vary in size
    - Different representation in files and screen than internal representation
    - 1, 34, –890 are all `int`s, but have different sizes on disk (when written as text)
  - Sequential access with fprintf and fscanf is not usually used to update records

Random-Access Files

- Random access files
  - Access individual records without searching through other records
  - Instant access to records in a file
  - Data can be inserted without destroying other data
  - Data previously stored can be updated or deleted without overwriting
- Implemented using fixed length records
  - Sequential files do not have fixed length records

C’s view of a random-access file



Creating a Random-Access File

- Data in random access files
  - Unformatted (stored as "raw bytes")
    - All data of the same type (`int`s, for example) uses the same amount of memory
    - All records of the same type have a fixed length. Hence, the exact location of a record relative to the beginning of the file can be calculated as a function of the record key.
    - Data not human readable

Creating a Random-Access File

- Unformatted I/O functions
  - `fwrite`
    - Transfer bytes from a location in memory to a file
  - `fread`
    - Transfer bytes from a file to a location in memory
  - Example:

    ```
    fwrite( &number, sizeof( int ), 1, myPtr );
    ```

    - `&number` – Location to transfer bytes from
    - `sizeof( int )` – Number of bytes to transfer
    - `1` – For arrays, number of elements to transfer
      - In this case, "one element" of an array is being transferred
    - `myPtr` – File to transfer to or from

Creating a Random-Access File

- Writing `structs`

```
fwrite( &myObject, sizeof (struct myStruct), 1, myPtr );
```

  - `sizeof` – returns size in bytes of object in parentheses
- To write several array elements
  - Pointer to array as first argument
  - Number of elements to write as third argument

Creating a Random-Access File: Example

```
1  /* Fig. 11.11: fig11_11.c
2     Creating a random-access file sequentially */
3  #include <stdio.h>
4
5  /* clientData structure definition */
6  struct clientData {
7     int acctNum;           /* account number */
8     char lastName[ 15 ];   /* account last name */
9     char firstName[ 10 ]; /* account first name */
10    double balance;        /* account balance */
11 }; /* end structure clientData */
12
13 int main( void )
14 {
15    int i; /* counter used to count from 1-100 */
16
17    /* create clientData with default information */
18    struct clientData blankClient = { 0, "", "", 0.0 };
19
```

Creating a Random-Access File: Example

```
20    FILE *cfPtr; /* credit.dat file pointer */
21
22    /* fopen opens the file; exits if file cannot be opened */
23    if ( ( cfPtr = fopen( "credit.dat", "wb" ) ) == NULL ) {
24       printf( "File could not be opened.\n" );
25    } /* end if */
26    else {
27
28       /* output 100 blank records to file */
29       for ( i = 1; i <= 100; i++ ) {
30          fwrite( &blankClient, sizeof( struct clientData ), 1, cfPtr );
31       } /* end for */
32
33       fclose ( cfPtr ); /* fclose closes the file */
34    } /* end else */
35
36    return 0; /* indicates successful termination */
37
38 } /* end main */
```

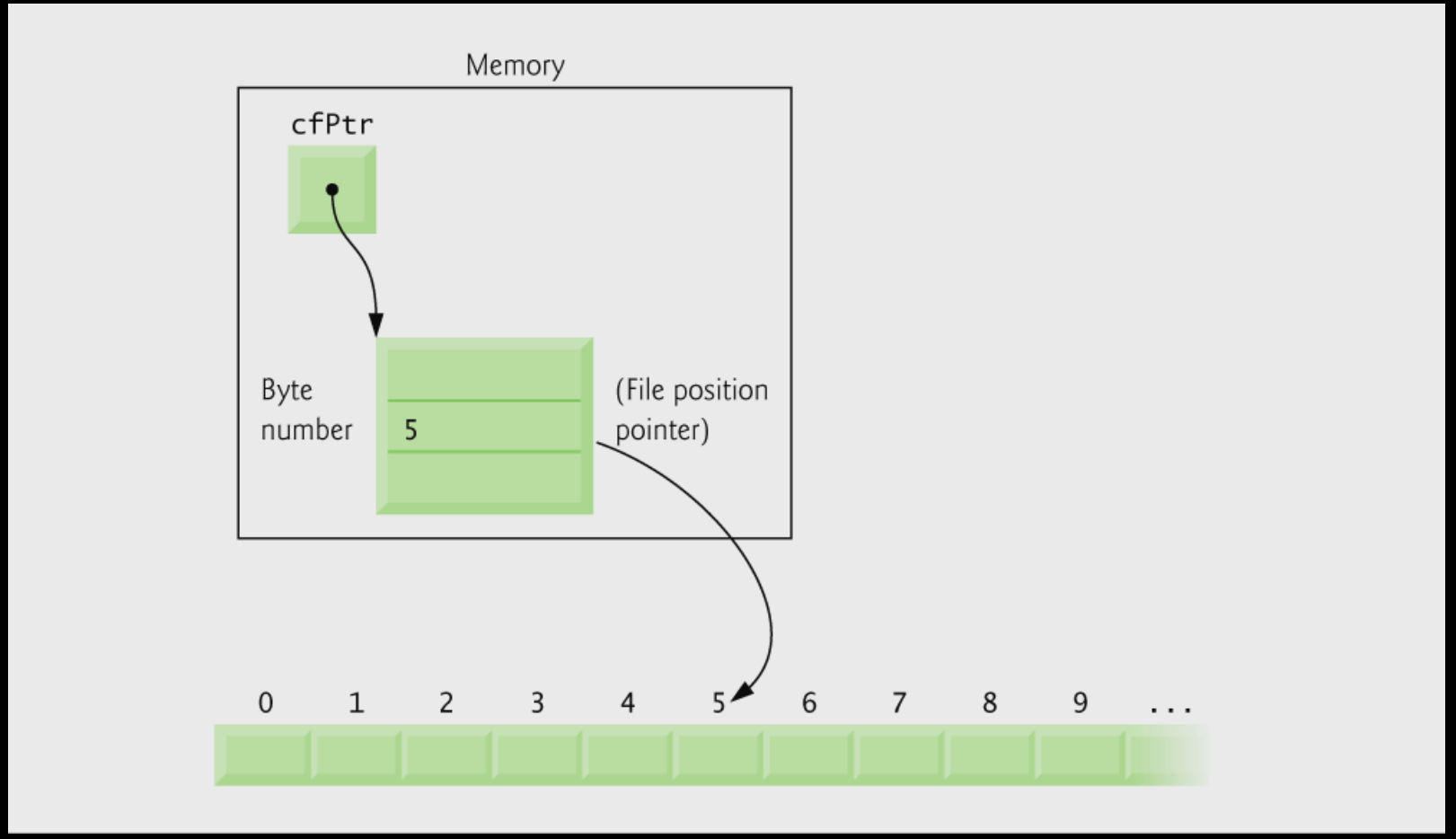
fopen function opens a file; **wb** argument means the file is opened for writing in binary mode

fwrite transfers bytes into a random-access file

Writing Data Randomly to a Random-Access File

- `fseek`
  - Sets file position pointer to a specific position
  - `fseek(` *pointer, offset, symbolic_constant* `);`
    - *pointer* – pointer to file
    - *offset* – file position pointer (0 is first location)
    - *symbolic_constant* – specifies where in file we are reading from
      - SEEK_SET – seek starts at beginning of file
      - SEEK_CUR – seek starts at current location in file
      - SEEK_END – seek starts at end of file
  - Function fseek returns a nonzero value if the seek operation cannot be performed.

File position pointer indicating an offset of 5 bytes from the beginning of the file



Writing to a Random-Access File: Example

```c
/* Fig. 11.12: fig11_12.c
   Writing to a random access file */
#include <stdio.h>

/* clientData structure definition */
struct clientData {
   int acctNum;           /* account number */
   char lastName[ 15 ];   /* account last name */
   char firstName[ 10 ]; /* account first name */
   double balance;        /* account balance */
}; /* end structure clientData */

int main( void )
{
   FILE *cfPtr; /* credit.dat file pointer */

   /* create clientData with default information */
   struct clientData client = { 0, "", "", 0.0 };

   /* fopen opens the file; exits if file cannot be opened */
   if ( ( cfPtr = fopen( "credit.dat", "rb+" ) ) == NULL ) {
      printf( "File could not be opened.\n" );
   } /* end if */
   else {

      /* require user to specify account number */
      printf( "Enter account number"
         " ( 1 to 100, 0 to end input )\n? " );
      scanf( "%d", &client.acctNum );

```

Writing to a Random-Access File: Example

```
31       /* user enters information, which is copied into file */
32       while ( client.acctNum != 0 ) {
33
34          /* user enters last name, first name and balance */
35          printf( "Enter lastname, firstname, balance\n? " );
36
37          /* set record lastName, firstName and balance value */
38          fscanf( stdin, "%s%s%lf", client.lastName,
39             client.firstName, &client.balance );
40
41          /* seek position in file to user-specified rec
42          fseek( cfPtr, ( client.acctNum - 1 ) *
43             sizeof( struct clientData ), SEEK_SET );
44
45          /* write user-specified information in file */
46          fwrite( &client, sizeof( struct clientData ), 1, cfPtr );
47
48          /* enable user to input another account number */
49          printf( "Enter account number\n? " );
50          scanf( "%d", &client.acctNum );
51       } /* end while */
52
53       fclose( cfPtr ); /* fclose closes the file */
54    } /* end else */
55
56    return 0; /* indicates successful termination */
57
58 } /* end main */
```

`fseek` searches for a specific location in the random-access file

Writing to a Random-Access File: Example

```
Enter account number ( 1 to 100, 0 to end input )
? 37
Enter lastname, firstname, balance
? Barker Doug 0.00
Enter account number
? 29
Enter lastname, firstname, balance
? Brown Nancy -24.54
Enter account number
? 96
Enter lastname, firstname, balance
? Stone Sam 34.98
Enter account number
? 88
Enter lastname, firstname, balance
? Smith Dave 258.34
Enter account number
? 33
Enter lastname, firstname, balance
? Dunn Stacey 314.33
Enter account number
? 0
```

Reading Data from a Random-Access File

- `fread`
  - Reads a specified number of bytes from a file into memory

```
fread( &client, sizeof (struct clientData), 1, myPtr );
```

  - Can read several fixed-size array elements
    - Provide pointer to array
    - Indicate number of elements to read
  - To read multiple elements, specify in third argument

Reading a Random-Access File: Example

```c
/* Fig. 11.15: fig11_15.c
   Reading a random access file sequentially */
#include <stdio.h>

/* clientData structure definition */
struct clientData {
   int acctNum;           /* account number */
   char lastName[ 15 ];   /* account last name */
   char firstName[ 10 ]; /* account first name */
   double balance;        /* account balance */
}; /* end structure clientData */

int main( void )
{
   FILE *cfPtr; /* credit.dat file pointer */

   /* create clientData with default information */
   struct clientData client = { 0, "", "", 0.0 };

   /* fopen opens the file; exits if file cannot be opened */
   if ( ( cfPtr = fopen( "credit.dat", "rb" ) ) == NULL ) {
      printf( "File could not be opened.\n" );
   } /* end if */
```

Reading a Random-Access File: Example

```
24    else {
25       printf( "%-6s%-16s%-11s%10s\n", "Acct", "Last Name",
26          "First Name", "Balance" );
27
28       /* read all records from file (until eof) */
29       while ( !feof( cfPtr ) ) {
30          fread( &client, sizeof( struct clientData ), 1, cfPtr );
31
32          /* display record */
33          if ( client.acctNum != 0 ) {
34             printf( "%-6d%-16s%-11s%10.2f\n",
35                client.acctNum, client.lastName,
36                client.firstName, client.balance );
37          } /* end if */
38
39       } /* end while */
40
41       fclose( cfPtr ); /* fclose closes the file */
42    } /* end else */
43
44    return 0; /* indicates successful termination */
45
46 } /* end main */
```

`fread` reads bytes from a random-access file to a location in memory

```
Acct  Last Name       First Name    Balance
29    Brown           Nancy          -24.54
33    Dunn            Stacey         314.33
37    Barker          Doug             0.00
88    Smith           Dave           258.34
96    Stone           Sam             34.98
```